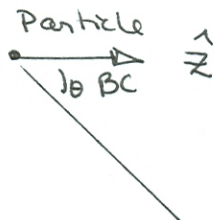


lol6



Emit photons $\frac{dN}{d\Omega'} = f(\theta', \varphi')$

Show that $\cos\theta' = \frac{\cos\theta - \beta}{1 - \beta\cos\theta}$

$$x'^{\mu} = (w'/c, \vec{k}')$$

$$w = kc$$

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu}$$

$$w'/c = \gamma w/c + \gamma\beta k_z$$

$$w'/c = k'$$

$$k'_z = -\gamma\beta w/c + \gamma k_z$$

$$\cos\theta' = \frac{k'_z}{k'}$$

$$k' = \gamma k - \gamma\beta k \cos\theta$$

$$k' \cos\theta' = \gamma k (\cos\theta - \beta)$$

$$\cos\theta' = \frac{\gamma k (\cos\theta - \beta)}{\gamma k (1 - \beta\cos\theta)} = \frac{(\cos\theta - \beta)}{(1 - \beta\cos\theta)}$$

$$\frac{dN}{d\Omega'} = f(\theta', \varphi')$$

need to change variables to θ and φ
note $d\varphi' = d\varphi$

$$\frac{d}{d\theta}(\cos\theta') = -\sin\theta' \frac{d\theta'}{d\theta} = \frac{-\sin\theta}{1 - \beta\cos\theta} - \frac{(\cos\theta - \beta)\beta\sin\theta}{(1 - \beta\cos\theta)^2}$$

$$\begin{aligned} \sin\theta' d\theta' &= \frac{\sin\theta(1 - \beta\cos\theta) + (\cos\theta - \beta)\beta\sin\theta}{(1 - \beta\cos\theta)^2} d\theta = \frac{\sin\theta - \beta^2\sin\theta}{(1 - \beta\cos\theta)^2} d\theta \\ &= \frac{1}{\gamma^2} \frac{\sin\theta}{(1 - \beta\cos\theta)^2} d\theta \end{aligned}$$

$$\frac{dN}{d\Omega} = \frac{dN}{\sin\theta d\theta d\varphi} = \frac{1}{\gamma^2(1 - \beta\cos\theta)^2} \frac{dN}{\sin\theta' d\theta' d\varphi'}$$

$$= \frac{f(\theta', \varphi')}{\gamma^2(1 - \beta\cos\theta)^2}$$